# Recent trends in GPU chips

One-page demo and test reference | Prepared 21 October 2026 | Selected 2024-2025 product examples, not an exhaustive market survey

Modern graphics processing units increasingly serve as AI compute engines as well as graphics processors. The main changes are larger high-bandwidth memory, specialized matrix arithmetic, multiple compute dies, and faster connections between accelerators. For language-model workloads, the practical question is not simply which chip has more peak operations: it is whether the model fits, meets response-time targets, and remains affordable at the expected request rate. This reference separates hardware facts from workload-dependent conclusions.

| Concrete example | Memory per accelerator | Published memory bandwidth |
|---|---|---|
| NVIDIA H200 Tensor Core GPU [1] | 141 GB HBM3e | 4.8 TB/s |
| AMD Instinct MI350X GPU [2] | 288 GB HBM3E | 8 TB/s |

Vendor specifications, not measured application results. Memory capacity and bandwidth alone do not establish which product is faster for a particular model.

## 1. Memory capacity is a deployment constraint

High-bandwidth memory, or HBM, sits close to the processor and supplies model weights, activations, and attention-cache data. A 70-billion-parameter model needs about 140 GB for weights alone at two bytes per parameter; four-bit weights require about 35 GB before scales, metadata, runtime buffers, and the key-value cache. These are arithmetic estimates, not deployment guarantees. Longer context windows and more concurrent users increase cache requirements. The H200 and MI350X examples illustrate why both capacity and memory headroom must be checked before choosing a server.

## 2. Bandwidth matters alongside compute

Autoregressive text generation repeatedly reads weights and cache data while producing tokens. At small batch sizes, memory traffic can limit throughput before peak arithmetic capacity is reached. HBM bandwidth is therefore a useful screening metric, but published terabytes per second describe a hardware limit, not end-to-end tokens per second. Prefill, the initial processing of a prompt, and decode, the subsequent generation of tokens, stress hardware differently. Benchmark both phases with realistic input lengths, output lengths, batch sizes, and concurrent requests. Do not treat a bandwidth ratio as a guaranteed application speedup.

## 3. Lower precision reduces resource demand

FP8 and newer FP4 execution paths reduce the number of bits used for selected tensor operations. NVIDIA Blackwell includes a second-generation Transformer Engine with FP4 support [3]. Less precision can lower memory traffic and improve supported matrix throughput, but numeric formats are not interchangeable. Software kernels, quantization scales, and model calibration determine whether an application benefits. Check factual accuracy, answer format, and task-specific quality after quantization. Retain higher precision where needed; a faster model that fails the quality gate is not a successful replacement.

## 4. Chiplets and interconnects scale the system

Blackwell combines two large compute dies into one GPU using a high-bandwidth die-to-die connection [3]. This multi-die approach helps expand compute beyond a single die's manufacturing limits. At server scale, accelerator links such as NVLink and AMD Infinity Fabric carry data between devices. These links differ from a GPU's local HBM interface. Splitting model weights across accelerators introduces communication and synchronization, so doubling device count does not guarantee twice the useful throughput. Compare an entire configured server, including its topology and software, rather than isolated chip specifications.

## 5. Power, cooling, and software affect value

Dense accelerator servers require adequate electrical supply, heat removal, and physical capacity. Chip power ratings are not the same as whole-server energy use: CPUs, memory, networking, fans, and cooling infrastructure also consume power. Evaluate energy per completed task and cost per accepted response instead of peak compute alone. Software compatibility is equally important. NVIDIA CUDA and AMD ROCm have different supported frameworks, kernels, and operational requirements. Confirm that the intended model, quantization method, inference engine, and monitoring tools work on the exact target configuration before committing to a migration.

## 6. Use a repeatable acceptance test

Hold the model, prompt set, context length, output limit, and quality rubric constant. Record device count, memory configuration, numeric precision, batch size, and software versions. Measure time to first token, generation throughput, p95 end-to-end latency, peak memory, and total operating cost. Include short and long prompts plus realistic concurrency. Warm up the system before measuring and report the number of requests. Repeat the run rather than relying on a single observation. Choose the least costly configuration that satisfies the same quality and service targets; there is no universal GPU winner.

## Demo questions answered by this document

How much HBM does the H200 provide? Which listed GPU has 288 GB of memory? Why can decode be bandwidth-bound? What is the weight-only four-bit estimate for a 70B model? Why does adding GPUs not guarantee linear speedup? Which latency and quality checks should remain fixed when comparing accelerator configurations?

**Sources:** [1] NVIDIA H200 product specifications: https://www.nvidia.com/en-us/data-center/h200/
[2] AMD Instinct MI350X specifications: https://www.amd.com/en/products/accelerators/instinct/mi350/mi350x.html
[3] NVIDIA Blackwell architecture: https://www.nvidia.com/en-us/data-center/technologies/blackwell-architecture/
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